

Jarrett D. Phillips, Ph.D.
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Department of Integrative Biology
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ACADEMIC APPOINTMENTS

Adjunct Professor, School of Computer Science 2023–Present
University of Guelph
Affiliated Faculty, One Health Institute 2026
University of Guelph

EDUCATION

Ph.D. in Computational Sciences, University of Guelph 2016–2022
Co-Advisors: Dr. Daniel Gillis and Dr. Robert Hanner
Advisory Committee Members: Dr. Deborah Stacey and Dr. Graham Taylor
Dissertation: A Novel Statistical Framework for Assessment of Intraspecific Haplotype Sampling
Completeness: Implications for DNA Barcode Gap Estimation
Master of Bioinformatics, University of Guelph 2013–2014
Co-Advisors: Dr. Robert Hanner and Dr. Daniel Ashlock
Major Paper: Assessing DNA Barcode Haplotype Sampling Diversity in the Ray-finned Fishes (Chordata: Actinopterygii)
BSc. (Hons.) in Biological Science, University of Guelph 2009–2013
Coursework in bioinformatics, ecology, evolutionary biology, comparative animal physiology, genetics, mathematics, and statistics

RESEARCH EXPERIENCE

Postdoctoral Fellow 2023–Present
University of Guelph
GBADs Informatics Team
Stacey Lab, School of Computer Science
Supervisor: Dr. Deborah Stacey
Postdoctoral Fellow 2023–Present
University of Guelph
Gillis Lab, School of Computer Science
Hanner Lab, Department of Integrative Biology
Supervisors: Drs. Daniel Gillis and Robert Hanner
Postdoctoral Fellow 2022
University of Guelph
Hanner Lab, Department of Integrative Biology
Supervisor: Dr. Robert Hanner

SOFTWARE DEVELOPMENT

GBADs ModelBuilder · R Shiny web application
HACSim (Haplotype Accumulation Curve Simulator) · R package · R Shiny web application
Frequentist/Bayesian DNA barcode gap coalescent-informed estimators · R scripts
RulesTools · R package
Seafood Fraud Modelling App · Python web application
VLF (Very Low Frequency) · R package

TEACHING EXPERIENCE

Course Instructor University of Guelph CIS*1910 – Discrete Structures in Computing I	2025
Graduate Teaching Assistant (GTA) University of Guelph CIS*3130 – System Modelling and Simulation CIS*1910 – Discrete Structures in Computing I CIS*2460 – Modelling of Computer Systems	2016–2020 2020 2017 2016–2019

STUDENT SUPERVISION AND MENTORSHIP

* Indicates students under my direct mentorship/supervision
** Indicates students under my co-mentorship/co-supervision

Undergraduate

*Brendan Carl Rosario Undergraduate Student Volunteer · Supervised Machine Learning for eDNA spatiotemporal sampling	2025
*Richard Cui Summer Undergraduate Research Assistant (URA) · Dynamical modelling of seafood fraud in the supply chain	2025
*Nikolett Toth CIS*4900/4910 · Mining association rules for eDNA spatiotemporal sampling	2024-2025
*Nikolett Toth Summer Undergraduate Research Assistant (URA) · Mining association rules for eDNA spatiotemporal sampling	2024
*Fynn De Vuono-Fraser CIS*4900/4910 · Bayesian modelling of seafood fraud in the Canadian supply chain	2024
*Zaid Al-Gayyali Summer Undergraduate Research Assistant (URA) · Seafood Fraud Visualization Tool Shiny app	2023
*Fynn De Vuono-Fraser STAT*4600 · Bayesian modelling of seafood fraud in the Canadian supply chain	2023

*Navdeep Singh	2021
CIS*4900 · HACSim R Shiny web application	
*Scarlett Bootsma	2020–2021
CIS*4900/4910 · HACSim simulation study	
**Maya Persram	2020
Hanner Lab volunteer · R reporting ecological meta-analysis	
**Ashley Chen	2020
Hanner Lab volunteer · R reporting ecological meta-analysis	
**Olivia Friesen Kroeker	2020
Hanner Lab volunteer · R reporting ecological meta-analysis	
*Steven French	2018
CIS*4900/4910 · HACSim R package	
**Julia Harvie	2018–2019
MCB*4500/4510 · Data mining GenBank and BOLD	

Graduate

**Nathan Zeinstra	2024
IBIO*6070 · Bayesian habitat occupancy modelling of sea lamprey using eDNA	
**Amina Asif	2022
BINF*6999 · DNA barcode gap analysis of Canadian disease vectors and agricultural pests	
**Danielle St. Jean (withdrawn)	2018–2019
MSc. thesis (Mathematics) · DNA barcode sequence classification with machine learning	
**Christina Frigel	2018–2019
BINF*6999 · DNA barcode sequence classification with machine learning	
**Jiaojia (Paula) Yu	2018–2019
BINF*6999 · MDMAPR R Shiny app	
**Ankita Bhandari	2018
BINF*6999 · Data mining GenBank and BOLD	

ASSISTANTSHIPS, AWARDS, SCHOLARSHIPS, AND GRANTS¹

Completed

Food From Thought Advancing Research Impact (ARIF) Fund – Livestock Innovation Grant	2024
A Modular Decision- and Policy-Making Tool for Global Livestock Disease Burden Assessment	
University of Guelph	\$40000.00 CAD
Food From Thought Advancing Research Impact (ARIF) Fund	2023
Forecasting Seafood Fraud Occurrence in the Supply Chain Within Major Canadian Cities Using Bayesian Hierarchical Binary Logistic Time-Series Regression Modeling	
University of Guelph	\$30000.00 CAD
SoCS Travel Grant	2019
University of Guelph	\$1000.00 CAD
Arthur D. Latornell Graduate Travel Grant	2019
University of Guelph	\$500.00 CAD

Graduate Teaching Assistantships	2016–2020
University of Guelph	\$35000 CAD
Graduate Research Assistantships	2017–2019
University of Guelph	\$11000.00 CAD
CPES Graduate Dean’s Scholarship	2017
University of Guelph	\$3500.00 CAD
CPES Graduate Excellence Entrance (GEE) Scholarship	2016
University of Guelph	\$30000.00 CAD

To be Submitted

NSERC Discovery Grant	2026
University of Guelph	Amount \$TBD CAD
Computational Statistical Methods for Specimen Sampling Sufficiency and Species Genetic Diversity Assessment	
Advancing Research Impact Fund: “Quick Wins” Knowledge Mobilization Fund	2026
University of Guelph	\$20000 CAD
GBADs ModelBuilder as a Tool to Teach Livestock Population Dynamics and Disease Burden Risk Estimation	
NSERC Alliance Society Grant	2026
University of Guelph	\$2.5 million CAD
AI-Driven Biodiversity Restoration: Scalable Solutions for Ecosystem Monitoring and Conservation	
NSERC CREATE Grant	2026
University of Guelph	\$9.9 million CAD
AI-Driven Training of Next-Generation Conservation Scientists	

ACADEMIC SERVICE

Pathway to Increase Standards and Competency in eDNA Surveys (PISCeS) Conference	2026
University of Guelph	
• Conference organizer and volunteer	
39th Canadian Artificial Intelligence Conference	2026
Simon Fraser University	
• Reviewer for Long and Short Papers Track	
GBADs Informatics Technical Workshop	2023
University of Liverpool	
• Workshop organizer	
Pathway to Increase Standards and Competency in eDNA Surveys (PISCeS) Conference	2023
University of Guelph	
• Conference organizer and volunteer	
School of Computer Science (SoCS) Faculty Search Committee	2018
University of Guelph	
• Member of hiring panel for Associate Professor in Cybersecurity	
School of Computer Science (SoCS) Faculty Search Committee	2017–2018
University of Guelph	
• Member of hiring panel for 2-year Contractually Limited Assistant Professor in Cybersecurity	

ACADEMIC PEER REVIEW SERVICE

Ecology and Evolution, F1000 Research, Frontiers in Ecology and Evolution, Lifestyle Genomics, Mitochondrial DNA Part A, Molecular Ecology Resources, Molecular Biology Reports, Methods in Ecology and Evolution, Nature Communications

PROCEEDINGS

* Indicates students under my direct mentorship/supervision

Pathway to Increase Standards and Competency of eDNA Surveys (PICSeS) International Conference (oral presentation) University of Guelph, Canada Phillips, J.D. · Association rule mining for targeted spatiotemporal aquatic environmental DNA (eDNA) sampling	2026
Pathway to Increase Standards and Competency of eDNA Surveys (PICSeS) International Conference (poster presentation) University of Guelph, Canada *Toth, N., Antonie, M.L. and Phillips, J.D. · To correct or not to correct?: Assessing the multiple comparisons problem for association rule mining of environmental DNA (eDNA) detection survey datasets	2026
39th Canadian Artificial Intelligence Conference (short paper and student poster) Simon Fraser University, Canada * · Toth, N., Antonie, M.L. and Phillips, J.D. · To correct or not to correct?: Assessing the multiple comparisons problem for association rule mining of environmental DNA (eDNA) detection survey datasets.	2026
SoCS Undergraduate Summer Project Show and Tell (student oral presentation) University of Guelph, Canada *Cui, R.C., Yodzis, M.P., and Phillips, J.D. · The Importance in Design as a Computer Scientist	2025
CEPS Undergraduate Student Poster Day (student poster presentation) University of Guelph, Canada *Toth N. and Phillips, J.D. · Association Rule Mining of eDNA Datasets	2024
CBS Undergraduate Poster Session (student poster presentation) University of Guelph, Canada * Toth N. and Phillips, J.D. · Association Rule Mining of eDNA Datasets	2024
9th International Barcode of Life Conference (poster presentation) Estação das Docas, Belem, Brazil Phillips, J.D. · A Measure of the DNA Barcode Gap for Applied and Basic Research (<i>not attended</i>)	2024
GBADs Informatics Technical Workshop (oral presentation) University of Liverpool, England Phillips, J.D. · The GBADs R Package (and Why We Need It!)	2023
CEPS Student Research Day (student poster presentation) University of Guelph, Canada *De Vuono-Fraser, F.A. and Phillips, J.D. · Estimating Seafood Mislabelling Rates in Canada Using Bayesian Modelling	2023

Pathway to Increase Standards and Competency of eDNA Surveys (PICSeS)	2023
International Conference (poster presentation) University of Guelph, Canada	
Phillips, J.D. · Haplotype diversity reveals challenges and opportunities for developing targeted detection assays for <i>COI</i> in Canadian freshwater fishes	
8th International Barcode of Life Conference (oral presentation)	2019
NTNU University Museum and Norwegian Biodiversity Information Centre, Norway	
Phillips, J.D. · HACSim: Iterative extrapolation of haplotype accumulation curves for assessment of intraspecific COI DNA barcode sampling completeness	
Guelph BioMathematics and Statistics (BioM&S) Symposium	
Artificial Intelligence and Machine Learning in Biology (attended)	2019
University of Guelph, Canada	
CEPS Undergraduate Student Poster Day (student poster presentation)	2018
University of Guelph, Canada	
*French, S.H. and Phillips, J.D. · Estimating Sampling Size Using Haplotype Accumulation Curves and Semiparametric Models	
7th International Barcode of Life Conference (oral presentation)	2017
University of Johannesburg, South Africa	
Phillips, J.D. · Intraspecific sample size estimation for DNA barcoding: Are current sampling levels enough?	
6th International Barcode of Life Conference (poster presentation)	2015
University of Guelph, Canada	
Phillips, J.D. · An exploration of sufficient sampling effort to describe intraspecific DNA barcode haplotype diversity: examples from the ray-finned fishes (Chordata: Actinopterygii).	

REFEREED WORK

- * Indicates students under my direct mentorship/supervision
- ** Indicates students under my co-mentorship/co-supervision

Journal Articles

Published or Accepted

10. **Phillips, J.D.** and *De Vuono-Fraser, F.A. (2026). Statistical modelling of seafood fraud highlights uncertainties in products from Metro Vancouver, British Columbia, Canada: Revisiting Hu *et al.* (2018). *Journal of Food Science*, **91**: e71201. DOI: [10.1111/1750-3841.71201](https://doi.org/10.1111/1750-3841.71201)
9. **Phillips, J.D.**, Hubert, N., and Hanner, R.H. (2025). A Bayesian coalescent model of the DNA barcode gap. Authorea. DOI: [10.22541/au.174073683.37707806/v1](https://doi.org/10.22541/au.174073683.37707806/v1)
8. Raymond, K., Sobkowich, K.E., **Phillips, J.D.**, Nguyen, L., McKechnie, I., Mohideen, R.N., Fitzjohn, W., Szurkowski, M., Davidson, J., Rushton, J., Stacey, D.A., and Bernardo T.M. (2024). GBADs informatics strategy: User-centric tools, data quality, and model interoperability. *WOAH Scientific and Technical Review*, **43**: 96-107. DOI: [10.20506/rst.43.3522](https://doi.org/10.20506/rst.43.3522).
7. **Phillips, J.D.** and *De Vuono-Fraser, F.A. (2024). Statistical modelling of seafood fraud in the Canadian supply chain. bioRxiv. DOI: [10.1101/2024.02.05.578947](https://doi.org/10.1101/2024.02.05.578947).

6. **Phillips, J.D.**, Athey, T.B.T., Hanner, R.H., and McNicholas, P.D. (2023). VLF: An R package for the analysis of very low frequency variants in DNA sequences. *Biodiversity Data Journal*, e96480. DOI: [10.3897/BDJ.11.e98480](https://doi.org/10.3897/BDJ.11.e98480).
5. **Phillips, J.D.**, Gillis, D.J., and Hanner, R.H. (2022). Lack of statistical rigor in DNA barcoding likely invalidates the presence of a true species' barcode gap. *Frontiers in Ecology and Evolution*, 10: 859099. DOI: [10.3389/fevo.2022.859099](https://doi.org/10.3389/fevo.2022.859099).
4. D'Ercole, J., Dincă, V., Opler, P.A., Kondla, N.G., Schmidt, C.B., **Phillips, J.D.**, Robbins, R., Burns, J.M., Miller, S.E., Grishin, N., Zakharov, E.V., deWaard, J.R., Ratnasingham, S., and Hebert, P.D.N. (2021). A DNA barcode library for the butterflies of North America. *PeerJ*, 9: e11157. DOI: [10.7717/peerj.11157](https://doi.org/10.7717/peerj.11157).
3. **Phillips, J.D.**, *French, S.H., Hanner, R.H., and Gillis, D.J. (2020). HACSim: An R package to estimate intraspecific sample sizes for genetic diversity assessment using haplotype accumulation curves. *PeerJ Computer Science*, 6(192): 1-37. DOI: [10.7717/peerj-cs.243](https://doi.org/10.7717/peerj-cs.243).
2. **Phillips, J.D.**, Gillis, D.J., and Hanner, R.H. (2019). Incomplete estimates of genetic diversity within species: Implications for DNA barcoding. *Ecology and Evolution*, 9(5): 2996-3010. DOI: [10.1002/ece3.4757](https://doi.org/10.1002/ece3.4757).
1. **Phillips, J.D.**, Gwiazdowski, R.A., Ashlock, D., and Hanner, R. (2015). An exploration of sufficient sampling effort to describe intraspecific DNA barcode haplotype diversity: examples from the ray-finned fishes (Chordata: Actinopterygii). *DNA Barcodes*, 3: 66-73. DOI: [10.1515/dna-2015-0008](https://doi.org/10.1515/dna-2015-0008).

Submitted or Under Revision

1. **Phillips, J.D.** and *De Vuono-Fraser, F.A. Swimming in uncertainty: How proper statistical modelling can help expose seafood product mislabeling. Submitted to *CHANCE*.

To Be Submitted

10. Elliott, T.A., Diao, J., **Phillips, J.D.**, and Adamowicz, S.J. Benchmarking imputation methods for insect trait data across missingness mechanisms and trait types. Targeted to *Methods in Ecology and Evolution*.
9. **Phillips, J.D.**, Hubert, N., and Hanner, R.H. A Bayesian-informed coalescent model of the DNA barcode gap. Targeted to *Methods in Ecology and Evolution*.
8. **Phillips, J.D.**, *Cui, R.C., and Yodzis, M.P. Seafood market product fraud erodes both consumer trust and resource sustainability within the global supply chain. Targeted to *Proceedings of the Royal Society B: Biological Sciences*.
7. *Toth, N., Antonie, M.L., Hanner, R.H., Gillis, D.J., and **Phillips, J.D.** Mining association rules for targeted spatiotemporal aquatic environmental DNA (eDNA) sampling. Targeted to *Oikos*.
6. **Phillips, J.D.**, *Al-Gayyali, Z.B., *De Vuono-Fraser, F.A., Hanner, R.H. and Gillis, D.J. The Seafood Fraud Visualization Tool: An R Shiny web app to summarize, model, and visualize seafood mislabelling trends in the supply chain.
5. Morey, K.C., **Phillips, J.D.**, Loeza-Quintana, T., and Hanner, R.H. Haplotype diversity reveals challenges and opportunities for developing targeted detection assays for COI in Canadian freshwater fish. Targeted to *Environmental DNA*.
4. Young, R.G., **Persram, M., **Friesen, O., **Chen, A., **Yu, J., **Phillips, J.D.**, and Hanner, R.H. Incomplete and irregular reporting of the R statistical and computing environment highlights the need for citation guidelines to support scientific reproducibility.
3. **Phillips, J.D.**, *Singh, N., Hanner, R.H., and Gillis, D.J. The HACSim R Shiny app: A web interface to estimate specimen sampling sufficiency for species genetic diversity assessment with DNA sequence data.

2. D'Ercole, J., Dapporto, L., **Phillips, J.D.**, Dincă, V.E., Vila, R., Talavera, G., and Hebert, P.D.N. Macrogenetics of North American butterflies—The impact of Quaternary climatic fluctuations. Targeted to *PNAS*.

1. **Phillips, J.D.**, *Bootsma, S.E., Hanner, R.H., and Gillis, D.J. Solving the genetic specimen sample size problem with a local search optimization algorithm. Targeted to *Methods in Ecology and Evolution*.

Book Chapters

Published or Accepted

2. **Phillips, J.D.**, Griswold, C.K., Young, R.G., Hubert, N., and Hanner, R.H. (2024). A Measure of the DNA Barcode Gap for Applied and Basic Research. In: DeSalle, R. (eds) DNA Barcoding. Methods in Molecular Biology, vol 2744. Humana, New York, NY.

URL: https://link.springer.com/protocol/10.1007/978-1-0716-3581-0_24

1. Hubert, N., **Phillips, J.D.**, and Hanner, R.H. (2024). Delimiting Species with Single-Locus DNA Sequences. In: DeSalle, R. (eds) DNA Barcoding. Methods in Molecular Biology, vol 2744. Humana, New York, NY. URL: https://link.springer.com/protocol/10.1007/978-1-0716-3581-0_3

Conference Proceedings

Published or Accepted

5. *Toth, N., Antonie, M.L., and **Phillips, J.D.** To correct or not to correct?: Assessing the multiple comparisons problem for association rule mining of environmental DNA (eDNA) detection survey datasets. *39th Canadian Conference on Artificial Intelligence*, **318**: 1012-1019. Short paper and poster.

4. Morey, K., Loeza-Quintana, T., **Phillips, J.**, and Hanner R. (2023). Haplotype diversity reveals challenges and opportunities for developing targeted detection assays for *COI* in Canadian freshwater fishes. Pathways to Increase Standards and Competency in eDNA Surveys (PISCeS) Conference. Poster.

3. **Phillips, J.D.**, Gillis, D., and Hanner, R. (2019). HACSim: Iterative extrapolation of haplotype accumulation curves for assessment of intraspecific COI DNA barcode sampling completeness. Scientific abstracts from the 8th International Barcode of Life Conference, Trondheim, Norway (ed. Torbjørn Ekrem), *Genome*, 62(6): 349-453. Oral presentation.

2. **Phillips, J.D.**, Gillis, D., and Hanner, R. (2017). Intraspecific sample size estimation for DNA barcoding: Are current sampling levels enough? Scientific abstracts from the 7th International Barcode of Life Conference, Johannesburg, South Africa (ed. M. van der Bank), *Genome*, 60(11): 881-1019. Oral presentation.

1. **Phillips, J.D.**, Gwiazdowski, R.A., Ashlock, D., and Hanner, R. (2015). An exploration of sufficient sampling effort to describe intraspecific haplotype diversity in the ray-finned fishes (Chordata: Actinopterygii). Scientific abstracts from the 6th International Barcode of Life Conference, Guelph, ON., Canada (ed. S.J. Adamowicz), *Genome*, 58(5): 163-303. Poster.

NON-REFEREED WORK

* Indicates students under my direct mentorship/supervision

15. *Cui, R.C. and **Phillips, J.D.** (2026). Student-contributed poster for University of Guelph March Open House
14. *Toth, N. and **Phillips, J.D.** (2026). Student-contributed poster for University of Guelph March Open House
13. *Cui, R.C. and **Phillips, J.D.** (2025). The Importance in Design as a Computer Scientist. Student-contributed SoCS Show and Tell video URL: <https://github.com/RichaCui70/Dynamics-between-Seafood-Fraudsters-and-Buyers/blob/main/Summer%20Showcase%20-%20Richard%20Cui.mp4>
12. **Phillips, J.D.**, *De Vuono-Fraser, F.A., Gillis, D.J., and Hanner, R.H. (2024). Statistical modelling of seafood fraud. Whiteboard explainer video. URL: <https://www.youtube.com/watch?v=4jU8cU8M70U>
11. *Toth, N. and **Phillips, J.D.** (2024). eDNA Collection Gets a Tech Update. Student-contributed CEPS Research Highlights article. URL: <https://www.uoguelph.ca/ceps/news/2024/07/edna-collection-gets-tech-upgrade>
10. *Toth, N. and **Phillips, J.D.** (2024). Unravelling eDNA with Association Rule Mining. Contributed guest post to the Science Borealis-syndicated blog of Dr. Daniel Gillis. URL: <https://danielgillis.wordpress.com/2024/08/15/unravelling-edna-with-association-rule-mining/>
9. **Phillips, J.D.** (2024). Summer URA Position. Contributed guest post to the Science Borealis-syndicated blog of Dr. Daniel Gillis. URL: <https://danielgillis.wordpress.com/2024/02/07/summer-ura-position-2/>
8. **Phillips, J.D.** (2022). A Novel Statistical Framework for Assessment of Intraspecific Haplotype Sampling Completeness: Implications for DNA Barcode Gap Estimation. Ph.D. Thesis. URL: <https://atrium.lib.uoguelph.ca/items/8addfcc5-f21c-4691-89b7-c4db051892eb>
7. **Phillips, J.D.** (2022). Mind the Gap — The DNA Barcode Gap, That Is. Contributed CEPS Research Highlights article. URL: <https://www.uoguelph.ca/ceps/news/2022/08/mind-gap---dna-barcode-gap>
6. **Phillips, J.D.** (2020). Barcode Cracking. Contributed CEPS Research Highlights article. URL: <https://www.uoguelph.ca/ceps/news/2020/02/barcode-cracking>
5. **Phillips, J.D.** (2020). Protecting Biodiversity Through the Lens of Genetic Diversity. Contributed guest post to the Science Borealis-syndicated blog of Dr. Daniel Gillis. URL: <https://danielgillis.wordpress.com/2020/01/30/protecting-biodiversity-through-the-lens-of-genetic-diversity/>
4. **Phillips, J.D.** (2019). IBOL8 and the Midnight Sun. Contributed guest post to the Science Borealis-syndicated blog of Dr. Daniel Gillis. URL: <https://danielgillis.wordpress.com/2019/07/02/reflections-ibol8-and-the-midnight-sun/>
3. **Phillips, J.D.** (2017). The Big Five and IBOL7. Contributed guest post to the Science Borealis-syndicated blog of Dr. Daniel Gillis. URL: <https://danielgillis.wordpress.com/2017/12/06/reflections-the-big-five-and-ibol7/>
2. **Phillips, J.D.** (2016). Sample size estimation for DNA barcoding: Are current sampling levels enough? Contributed guest post to the DNA Barcoding Blog of Dr. Dirk Steinke. URL: <http://dna-barcoding.blogspot.com/2016/01/guest-post-sample-size-estimation-for.html>
1. **Phillips, J.D.** (2016). Sample size estimation for DNA barcoding of ray-finned fishes: Are current sampling levels enough? Contributed newsletter article to the Barcode Bulletin, 7(1). URL: <https://journal.lib.uoguelph.ca/index.php/ibol/issue/view/345>

VOLUNTEER EXPERIENCE

CEPS Student Research Connections Networking Night

2025

University of Guelph

- Connected with undergraduate students for summer URA project recruitment

CIS*3750 Wireframing session

2021-2025

University of Guelph

- Graded students on mobile app prototypes for various community partners using Qualtrics surveys

REFERENCES

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